POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 3012 | Semester III | Theory | 4 Credits

Advanced Surveying

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Civil Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

Course identity, Semester III teaching scheme, credits, CIE/ESE and examination mode were verified from the official SITTTR Kerala Civil Engineering Revision 2026 programme scheme. The official SITTTR Course 3012 detail endpoint was unavailable during preparation. With user approval, explanatory coverage uses a reliable public advanced-surveying curriculum and is not presented as recovered official Course 3012 detailed-syllabus wording.

Source treatment: The SITTTR programme scheme establishes the course metadata. The handbook groups conventional advanced-surveying topics into four study modules based on a public diploma curriculum covering theodolite, levelling, tacheometry, curves and total station. It is a transparent learning aid, not a reproduction of unavailable SITTTR detailed content.

Verified source note: The course detail link reached from the official programme scheme failed during source retrieval. The official programme-scheme source and approved public advanced-surveying curriculum are linked in References.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Advanced Surveying extends field-measurement practice from basic instruments to angular measurement, traversing, trigonometrical levelling, tacheometry, curve setting and electronic survey systems. The emphasis is on systematic setup, field-book discipline, calculations and error control.

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Prerequisite foundation

Basic Surveying (2012), trigonometry, coordinate geometry and careful field-recording practice.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Operate theodolite concepts for angle measurement and traversing.
2. Use trigonometrical levelling and tacheometry principles for distance/elevation work.
3. Calculate and set out simple circular curves.
4. Explain total-station setup, observation, data handling and field precautions.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain theodolite use, angle measurement and traverse computation.	Apply
CO2	Apply trigonometrical levelling and tacheometric methods to find distance and elevation.	Apply
CO3	Calculate elements and set out simple circular curves.	Apply
CO4	Explain total-station setup, survey workflow, data retrieval and error control.	Apply

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	Title	Official major topics	Hours	CO	Handbook section
1	Theodolite Surveying and Traverses	Theodolite parts, temporary adjustment, horizontal/vertical angles, bearings, traversing, closing error and balancing.	Approx. 16 hours	CO1	Module 1
2	Trigonometrical Levelling and Tacheometry	Elevation by vertical-angle observation; stadia/tangential methods; horizontal distance and reduced level calculations.	Approx. 15 hours	CO2	Module 2
3	Curves and Field Setting Out	Circular-curve terminology, radius/degree relation, curve elements, offsets, Rankine method and transition/vertical-curve purpose.	Approx. 14 hours	CO3	Module 3
4	Electronic Survey Systems and Total Station	Digital theodolite/EDM principles, total-station parts, centering/levelling/backsight, measurement, field recording, data retrieval and error control.	Approx. 15 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Theodolite Surveying and Traverses

Theodolite parts, temporary adjustment, horizontal/vertical angles, bearings, traversing, closing error and balancing.

CO1 · Approx. 16 hours

Trigonometrical Levelling and Tacheometry

Elevation by vertical-angle observation; stadia/tangential methods; horizontal distance and reduced level calculations.

CO2 · Approx. 15 hours

Curves and Field Setting Out

Circular-curve terminology, radius/degree relation, curve elements, offsets, Rankine method and transition/vertical-curve purpose.

CO3 · Approx. 14 hours

Electronic Survey Systems and Total Station

Digital theodolite/EDM principles, total-station parts, centering/levelling/backsight, measurement, field recording, data retrieval and error control.

CO4 · Approx. 15 hours

MODULE 1

Theodolite Surveying and Traverses

Theodolite parts, temporary adjustment, horizontal/vertical angles, bearings, traversing, closing error and balancing.

Approx. 16 hours **CO1** **Understand · Apply**

Learning goals

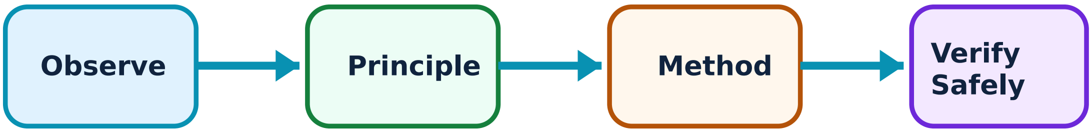
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Theodolite Surveying and Traverses: identify the system, state its principle, follow the correct method and verify the result safely.

1. Theodolite setup and temporary adjustments−

A theodolite measures horizontal and vertical angles only when it is centred over the station, levelled and focused correctly. Temporary adjustments establish reliable geometry between instrument, station and reference line.

Study and application method

Set tripod firmly, centre over station, level using foot screws, focus eyepiece/object and check bubble/centering again before observation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Set tripod firmly, centre over station, level using foot screws, focus eyepiece/object and check bubble/centering again before observation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not record angles before rechecking centering and levelling after changing instrument position.

2. Horizontal and vertical angle observation−

Angular observation uses a defined backsight, instrument reading and foresight. Repetition/reiteration and careful booking improve reliability. Vertical angles support height and slope calculations when referenced correctly.

Study and application method

State backsight and foresight, observe faces/sets as instructed, book both readings with station names and calculate the required angle using the stated convention.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State backsight and foresight, observe faces/sets as instructed, book both readings with station names and calculate the required angle using the stated convention.

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Common mistake / precaution: Never mix whole-circle bearing, reduced bearing and deflection-angle conventions in one computation.

3. Theodolite traversing and closing error−

A traverse is a connected series of survey lines. Latitude/departure computations and closing error reveal consistency of field observations; balancing distributes permissible correction according to an approved method.

Study and application method

Prepare a complete traverse table, retain sign conventions, calculate closure and record the selected balancing method before correcting coordinates.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Prepare a complete traverse table, retain sign conventions, calculate closure and record the selected balancing method before correcting coordinates.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A balanced traverse does not erase poor field practice; investigate large closure before accepting a plan.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Trigonometrical Levelling and Tacheometry

Elevation by vertical-angle observation; stadia/tangential methods; horizontal distance and reduced level calculations.

Approx. 15 hours | CO2 | Understand · Apply

Learning goals

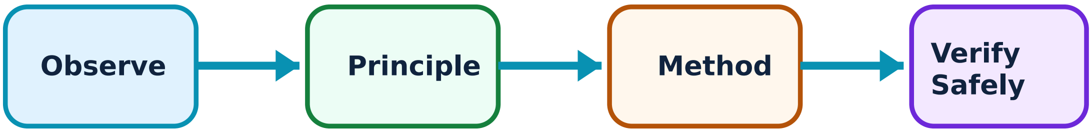
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Trigonometrical Levelling and Tacheometry: identify the system, state its principle, follow the correct method and verify the result safely.

1. Trigonometrical levelling−

Trigonometrical levelling determines relative elevation from measured vertical angle, distance, instrument height and target/staff reading. It is useful where direct levelling is difficult or objects are inaccessible.

Study and application method

Draw a height-of-instrument sketch, identify accessible/inaccessible base condition, apply the appropriate elevation relation and show every height correction with unit.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw a height-of-instrument sketch, identify accessible/inaccessible base condition, apply the appropriate elevation relation and show every height correction with unit.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Use a consistent vertical-angle reference and include instrument/target heights; omitting either causes a systematic error.

2. Stadia tacheometry−

Tacheometry obtains distance and elevation rapidly by observing staff intercept and vertical angle. Stadia constants and line-of-sight condition determine the calculation form.

Study and application method

Record upper, central and lower staff readings, compute intercept, identify horizontal or inclined sight case and calculate distance/RL with clearly stated constants.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Record upper, central and lower staff readings, compute intercept, identify horizontal or inclined sight case and calculate distance/RL with clearly stated constants.

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Common mistake / precaution: Staff must be held correctly and readings recorded in order; a swapped reading invalidates the intercept.

3. Tangential method and checks−

The tangential method uses two vertical-angle observations to a staff at known interval. It is an alternative measurement approach with its own field conditions and sensitivity to angle-reading error.

Study and application method

Sketch the geometry, enter both observed angles and staff interval, use the selected formula, then compare reasonableness with field distance/elevation context.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the geometry, enter both observed angles and staff interval, use the selected formula, then compare reasonableness with field distance/elevation context.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Small angle-reading mistakes can greatly affect results; repeat doubtful observations.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Curves and Field Setting Out

Circular-curve terminology, radius/degree relation, curve elements, offsets, Rankine method and transition/vertical-curve purpose.

Approx. 14 hours | CO3 | Understand · Apply

Learning goals

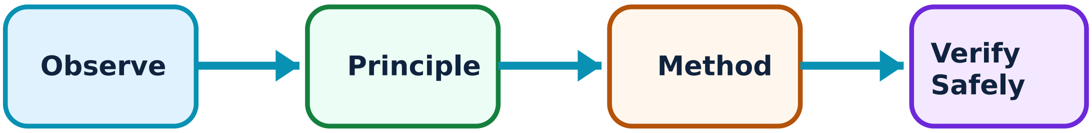
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Curves and Field Setting Out: identify the system, state its principle, follow the correct method and verify the result safely.

1. Simple circular curve elements−

A simple circular curve joins two tangents through a constant radius. Its tangent length, length of curve, central angle, chainage and deflection quantities are linked geometrically.

Study and application method

Draw tangent intersection, radius and curve centre; list given data, calculate one element at a time and retain chainage units consistently.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw tangent intersection, radius and curve centre; list given data, calculate one element at a time and retain chainage units consistently.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not confuse central angle with individual deflection angle or change degrees/radians without checking formula requirement.

2. Setting out by offsets and deflection angles−

Field setting methods convert calculated curve geometry into marked points. Offsets use measured perpendicular geometry; Rankine’s method uses cumulative deflection angles and chord lengths.

Study and application method

Prepare a setting-out table before field work with point, chainage/chord, angle/offset and check total; then record actual field observations separately.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Prepare a setting-out table before field work with point, chainage/chord, angle/offset and check total; then record actual field observations separately.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: [illegible] concept [illegible] [illegible] [illegible]. [illegible] diagram / circuit / formula / procedure [illegible] [illegible]. [illegible] result [illegible] application [illegible] [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Common mistake / precaution: Instrument orientation and cumulative versus incremental deflection must be verified before marking points.

3. Transition and vertical curves−

Transition curves provide gradual change in curvature between tangent and circular curve. Vertical curves provide gradual grade change in the vertical profile; both improve comfort, safety and layout continuity.

Study and application method

State purpose, location in alignment and one engineering benefit; use a labelled plan/profile sketch when explaining.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Do not treat horizontal and vertical curves as interchangeable; they are represented in different views and serve different geometry.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Electronic Survey Systems and Total Station

Digital theodolite/EDM principles, total-station parts, centering/levelling/backsight, measurement, field recording, data retrieval and error control.

Approx. 15 hours **CO4** **Understand · Apply**

Learning goals

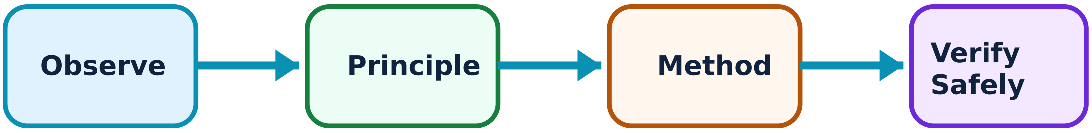
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Electronic Survey Systems and Total Station: identify the system, state its principle, follow the correct method and verify the result safely.

1. EDM and total-station principles−

A total station integrates angle measurement, electronic distance measurement and data storage. It supports coordinate collection, traversing, setting out and digital transfer when controlled procedures are followed.

Study and application method

Identify instrument components, state measurement inputs/outputs and draw a simple data-flow: setup → observation → storage → transfer → drawing/check.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify instrument components, state measurement inputs/outputs and draw a simple data-flow: setup → observation → storage → transfer → drawing/check.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Electronic measurement does not remove the need for correct prism setting, atmospheric configuration and independent checks.

2. Station setup and observation workflow−

Reliable total-station work requires stable tripod placement, centering, levelling, correct instrument/prism heights, known backsight or azimuth and properly named observations.

Study and application method

Follow a written field checklist, confirm station/backsight coordinates before measurement, record job name/point IDs and perform a check shot before leaving station.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Follow a written field checklist, confirm station/backsight coordinates before measurement, record job name/point IDs and perform a check shot before leaving station.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: A wrong point ID or backsight orientation can produce apparently precise but unusable coordinates.

3. Data retrieval, maintenance and error control−

Survey data must be transferred, checked and backed up before plan generation. Equipment care includes battery management, clean optics, protected transport and systematic identification/control of error sources.

Study and application method

Export data using approved format, compare independent checks, document coordinate system/datum and maintain a field log with file names and date.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Export data using approved format, compare independent checks, document coordinate system/datum and maintain a field log with file names and date.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Do not rely on one digital copy or alter raw data without preserving an original record.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Theodolite Surveying and Traverses

Use the concept flow to revise observation, principle, method and verification.

Module 2: Trigonometrical Levelling and Tacheometry

Use the concept flow to revise observation, principle, method and verification.

Module 3: Curves and Field Setting Out

Use the concept flow to revise observation, principle, method and verification.

Module 4: Electronic Survey Systems and Total Station

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Prepare a complete theodolite observation and traverse-book format; identify checks before balancing.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Solve a trigonometrical-levelling and a tacheometry example with labelled sketches and unit check.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Create a curve-setting table using an approved method and perform a desk-check of cumulative values.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Develop a total-station field checklist covering setup, point naming, backsight verification, data transfer and backup.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Theodolite Surveying and Traverses

Explain Theodolite setup and temporary adjustments and state one correct method, application or precaution.–

A theodolite measures horizontal and vertical angles only when it is centred over the station, levelled and focused correctly. Temporary adjustments establish reliable geometry between instrument, station and reference line.

Answer framework: Set tripod firmly, centre over station, level using foot screws, focus eyepiece/object and check bubble/centering again before observation.

Important point: Do not record angles before rechecking centering and levelling after changing instrument position.

Explain Horizontal and vertical angle observation and state one correct method, application or precaution.−

Angular observation uses a defined backsight, instrument reading and foresight. Repetition/reiteration and careful booking improve reliability. Vertical angles support height and slope calculations when referenced correctly.

Answer framework: State backsight and foresight, observe faces/sets as instructed, book both readings with station names and calculate the required angle using the stated convention.

Important point: Never mix whole-circle bearing, reduced bearing and deflection-angle conventions in one computation.

Explain Theodolite traversing and closing error and state one correct method, application or precaution.−

A traverse is a connected series of survey lines. Latitude/departure computations and closing error reveal consistency of field observations; balancing distributes permissible correction according to an approved method.

Answer framework: Prepare a complete traverse table, retain sign conventions, calculate closure and record the selected balancing method before correcting coordinates.

Important point: A balanced traverse does not erase poor field practice; investigate large closure before accepting a plan.

Module 2 — Trigonometrical Levelling and Tacheometry

Explain Trigonometrical levelling and state one correct method, application or precaution.−

Trigonometrical levelling determines relative elevation from measured vertical angle, distance, instrument height and target/staff reading. It is useful where direct levelling is difficult or objects are inaccessible.

Answer framework: Draw a height-of-instrument sketch, identify accessible/inaccessible base condition, apply the appropriate elevation relation and show every height correction with unit.

Important point: Use a consistent vertical-angle reference and include instrument/target heights; omitting either causes a systematic error.

Explain Stadia tacheometry and state one correct method, application or precaution.−

Tacheometry obtains distance and elevation rapidly by observing staff intercept and vertical angle. Stadia constants and line-of-sight condition determine the calculation form.

Answer framework: Record upper, central and lower staff readings, compute intercept, identify horizontal or inclined sight case and calculate distance/RL with clearly stated constants.

Important point: Staff must be held correctly and readings recorded in order; a swapped reading invalidates the intercept.

Explain Tangential method and checks and state one correct method, application or precaution.−

The tangential method uses two vertical-angle observations to a staff at known interval. It is an alternative measurement approach with its own field conditions and sensitivity to angle-reading error.

Answer framework: Sketch the geometry, enter both observed angles and staff interval, use the selected formula, then compare reasonableness with field distance/elevation context.

Important point: Small angle-reading mistakes can greatly affect results; repeat doubtful observations.

Module 3 — Curves and Field Setting Out

Explain Simple circular curve elements and state one correct method, application or precaution.−

A simple circular curve joins two tangents through a constant radius. Its tangent length, length of curve, central angle, chainage and deflection quantities are linked geometrically.

Answer framework: Draw tangent intersection, radius and curve centre; list given data, calculate one element at a time and retain chainage units consistently.

Important point: Do not confuse central angle with individual deflection angle or change degrees/radians without checking formula requirement.

Explain Setting out by offsets and deflection angles and state one correct method,

application or precaution.−

Field setting methods convert calculated curve geometry into marked points. Offsets use measured perpendicular geometry; Rankine's method uses cumulative deflection angles and chord lengths.

Answer framework: Prepare a setting-out table before field work with point, chainage/chord, angle/offset and check total; then record actual field observations separately.

Important point: Instrument orientation and cumulative versus incremental deflection must be verified before marking points.

Explain Transition and vertical curves and state one correct method, application or precaution.−

Transition curves provide gradual change in curvature between tangent and circular curve. Vertical curves provide gradual grade change in the vertical profile; both improve comfort, safety and layout continuity.

Answer framework: State purpose, location in alignment and one engineering benefit; use a labelled plan/profile sketch when explaining.

Important point: Do not treat horizontal and vertical curves as interchangeable; they are represented in different views and serve different geometry.

Module 4 — Electronic Survey Systems and Total Station

Explain EDM and total-station principles and state one correct method, application or precaution.−

A total station integrates angle measurement, electronic distance measurement and data storage. It supports coordinate collection, traversing, setting out and digital transfer when controlled procedures are followed.

Answer framework: Identify instrument components, state measurement inputs/outputs and draw a simple data-flow: setup → observation → storage → transfer → drawing/check.

Important point: Electronic measurement does not remove the need for correct prism setting, atmospheric configuration and independent checks.

Explain Station setup and observation workflow and state one correct method, application or precaution.−

Reliable total-station work requires stable tripod placement, centering, levelling, correct instrument/prism heights, known backsight or azimuth and properly named observations.

Answer framework: Follow a written field checklist, confirm station/backsight coordinates before measurement, record job name/point IDs and perform a check shot before leaving station.

Important point: A wrong point ID or backsight orientation can produce apparently precise but unusable coordinates.

Explain Data retrieval, maintenance and error control and state one correct method, application or precaution.−

Survey data must be transferred, checked and backed up before plan generation. Equipment care includes battery management, clean optics, protected transport and systematic identification/control of error sources.

Answer framework: Export data using approved format, compare independent checks, document coordinate system/datum and maintain a field log with file names and date.

Important point: Do not rely on one digital copy or alter raw data without preserving an original record.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Theodolite Surveying and Traverses.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Trigonometrical Levelling and Tacheometry.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Curves and Field Setting Out.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Electronic Survey Systems and Total Station.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.

2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Theodolite parts, temporary adjustment, horizontal/vertical angles, bearings, traversing, closing error and balancing..
2. Develop a complete answer or practical plan for: Elevation by vertical-angle observation; stadia/tangential methods; horizontal distance and reduced level calculations..
3. Develop a complete answer or practical plan for: Circular-curve terminology, radius/degree relation, curve elements, offsets, Rankine method and transition/vertical-curve purpose..
4. Develop a complete answer or practical plan for: Digital theodolite/EDM principles, total-station parts, centering/levelling/backsight, measurement, field recording, data retrieval and error control..

LAST-MILE REVISION

Quick Revision

Module 1: Theodolite Surveying and Traverses

Theodolite parts, temporary adjustment, horizontal/vertical angles, bearings, traversing, closing error and balancing.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Trigonometrical Levelling and Tacheometry

Elevation by vertical-angle observation; stadia/tangential methods; horizontal distance and reduced level calculations.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Curves and Field Setting Out

Circular-curve terminology, radius/degree relation, curve elements, offsets, Rankine method and transition/vertical-curve purpose.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Electronic Survey Systems and Total Station

Digital theodolite/EDM principles, total-station parts, centering/levelling/backsight, measurement, field recording, data retrieval and error control.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Theodolite setup and temporary adjustments	A theodolite measures horizontal and vertical angles only when it is centred over the station, levelled and focused correctly.
Horizontal and vertical angle observation	Angular observation uses a defined backsight, instrument reading and foresight.
Theodolite traversing and closing error	A traverse is a connected series of survey lines.
Trigonometrical levelling	Trigonometrical levelling determines relative elevation from measured vertical angle, distance, instrument height and target/staff reading.
Stadia tacheometry	Tacheometry obtains distance and elevation rapidly by observing staff intercept and vertical angle.
Tangential method and checks	The tangential method uses two vertical-angle observations to a staff at known interval.
Simple circular curve elements	A simple circular curve joins two tangents through a constant radius.
Setting out by offsets and deflection angles	Field setting methods convert calculated curve geometry into marked points.
Transition and vertical curves	Transition curves provide gradual change in curvature between tangent and circular curve.
EDM and total-station principles	A total station integrates angle measurement, electronic distance measurement and data storage.
Station setup and observation workflow	Reliable total-station work requires stable tripod placement, centering, levelling, correct instrument/prism heights, known backsight or azimuth and properly named observations.
Data retrieval, maintenance and error control	Survey data must be transferred, checked and backed up before plan generation.

LEARNING RESOURCES

References

Recommended advanced-surveying references

1. B. C. Punmia, Surveying Volumes I and II.
2. S. K. Duggal, Surveying Volumes I and II.
3. T. P. Kanetkar and S. V. Kulkarni, Surveying and Levelling.

4. SITTTR Kerala, Diploma Curriculum Revision 2026 Civil Engineering Programme Scheme (Course 3012 metadata).

Verified official and public curriculum sources

- https://www.sitttrkerala.ac.in/index.php?r=site%2Fdiploma-syllabus-courses&prog=CE&scheme=REV2026
- https://www.gtu.ac.in/Syllabus/New_Diploma/sem-4/Pdf/3340602.pdf

The SITTTR programme scheme verifies Course 3012 metadata. The public diploma curriculum supplies explanatory coverage while the official Course 3012 detail endpoint is unavailable; confirm all field procedures with departmental instructions and instrument manuals.